

Report On Artificial Intelligence

1. What is Artificial Intelligence?

Artificial Intelligence (AI) is a type of technology, which makes computers perform tasks, usually done by humans. Such tasks may involve learning, decision-making, problem-solving, language usage, image recognition, etc. Today, artificial intelligence has become an inherent part of our everyday life as we use it when we navigate through Google Maps, talk to chatbots, or receive recommendations from Netflix, etc.

Artificial intelligence enables people to work more productively and effectively as it solves complex problems within the shortest period. With the development of technology, today artificial intelligence becomes increasingly popular in healthcare, education, banking, transport, and other fields.

- Artificial intelligence allows performing tasks, which involve learning, analysis, and decision-making.
- Used in chatbots, voice assistants, recommendation systems, etc.
- Helps save time and become more efficient.
- Becomes important in different fields.

2. Difference Between AI, Machine Learning, Generative AI, and Agentic AI.

Artificial intelligence is a broad notion, which includes multiple approaches aimed at building intelligent algorithms. Below, let us compare four types of AI.

Machine learning involves teaching computers how to operate in a certain way by learning from data and experience. In turn, generative AI creates different types of content including texts, images, videos, and even codes. Agentic AI is the one, which operates independently as it plans, decides, and performs actions required to reach a goal.

Artificial Intelligence (AI):

- Makes machines perform certain intelligent tasks.
- Parent technology of other AI systems.

Machine Learning (ML):

- Learns and grows over time.
- Develops autonomously, meaning no programming is needed.

Generative AI:

- Creates something new.
- Produces text, images, videos, and code.

Agentic AI:

- Performs independently.
- Plans and takes actions to achieve goals.

3. Explain Any Three Open-Source AI Models Available on Hugging Face.

i. Llama 3:

It is a language model created by Meta that is capable of generating responses, making conclusions and analyses, providing coding solutions, and answering questions. Llama 3 is widely used by many developers and professionals to build AI-powered applications and chatbots.

- Language model created by Meta.
- Open source tool.
- Used for writing articles and codes.

ii. Mistral 7B:

It is a high-performance AI model, which is known to provide quality results while using less computing power. Thus, it is a good solution for creating a number of projects powered by AI technology.

- Very efficient.
- Can perform well on the limited set of computing power.
- Widely applied in different projects.

iii. Stable Diffusion:

Stable diffusion is a widely recognized AI model that creates images according to textual input from users. Users can create various artworks, graphics, and design elements simply by providing the text description of their idea.

- Generates images from textual inputs.
- Allows creating artworks.
- Open-source solution.

4. Difference Between Traditional Programming and AI-Assisted Development.

When using traditional programming, developers write instructions that a computer will use to perform a certain operation. Therefore, the computer has limitations as it cannot do anything that a developer has not included in the instruction. Traditional programming is very time-consuming and requires a lot of work.

When using AI-assisted development, artificial intelligence helps developers by suggesting codes, finding mistakes, writing documents, and performing any other routine tasks that are often performed when developing software products.

Traditional Programming:

- All instructions are written manually.
- Follows the logical rules and instructions.
- Takes more time and efforts.
- Output is based on what was written.

AI-Assisted Development:

- Helps to write and optimize codes.
- Finds mistakes and suggests their solutions.
- Saves time and efforts.
- Accelerates development.

5. Explain AI Hallucinations with One Example.

AI hallucination is a phenomenon where AI provides some information, which appears to be accurate but is incorrect. It happens since artificial intelligence uses machine learning to understand the task and provide the response. Therefore, it may invent information, names, or facts.

For instance, when asked about the list of novels created by a certain author, AI may suggest the name of the book that does not actually exist. Due to the fact that a response seems to be accurate, a person tends to believe it is true.

That is why any information provided by AI should be checked on additional sources before using it as an evidence in some papers.

- AI hallucination is the provision of incorrect information by AI.
- Information seems to be believable.
- Happens due to prediction of artificial intelligence.
- Limitation of AI.
- The information should be checked.

Example:

The creation of some book title by artificial intelligence is called hallucination.



NOTE: This is AI generated image.